2022 Prelim Data Analysis

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1.

The study is concerned with the performance of rVSVΔG-ZEBOV-GP vaccination in Congolese populations. The data set consists of 609 unique records of vaccinated participants. Each participant has three antibody titers at day 0, day 21, and month 6 postvaccination, which are measured multiple times by using FANG ELISA (we use median measurements). The data set also includes demographic variables (age, sex, education level, and marital status), timing since vaccination, whether the participant is a healthcare worker, and whether the participant has ever had a close contact to Ebola exposure. These variables are collected through interview questions. On the other hand, participants were not enrolled until after vaccination and the 30-min follow-up period to ensure official vaccination activities suffered no disruptions.

The authors used the antibody titers to check antibody response and antibody persistence, which can indicate the effect of the vaccination. Antibody response is defined as an antibody titer that had increased fourfold from visit 1 to visit 2 and was above four times the lower limit of quantification (LLOQ) at visit 2 (i.e., $> 267.84 = 4 \cdot 66.96$ EU/mL). Antibody persistence is also defined as a fourfold increase in antibody titer between visits 1 and 3 along with a visit 3 titer at least four times the LLOQ.

The authors observed that 87.2% of the returning participants had an antibody response, and 95.6% had an antibody persistence. Also, they observed an increase at each time point in the geometric mean of antibody titers. They argued that this observation is most likely a result of vaccine response but also potentially the product of natural exposure in the cohort, as the outbreak was ongoing during the study period. Despite the nonresponders and the breakthrough cases in Democratic Republic of the Congo (DRC), they suggested that most Congolese individuals exhibit a strong antibody response postvaccination.

The authors also examined demographics for associations with follow-up antibody titers using linear mixed models. Let y_{it} denote the antibody titer for each participant i and t-th measurement at day 21 (In the original study, antibody levels are measured multiple times). The univariate linear mixed models are:

$$\log_{10}(y_{it}) = \beta_0 + \beta x_i + u_i + \epsilon_{it}$$

where u_i is a random effect for participants, ϵ_{it} is an error term, and x_i is one of the predictors (sex, age category, education, marital status, health care worker status, and EVD exposure history). After fitting these models, the authors also fitted the same models with responses y_{it} at month 6. Moreover, they fitted the multivariate linear mixed models with predictors x_i which include all variables, for both responses at day 21 and month 6 respectively. Finally, they obtained $14 = 6 + 6 + 1 + 1$ models and reported the results (with exponential transformed coefficients) on Tables 3 and 4 in the paper.

From the fitted models, the authors observed female participants were more likely to have a higher antibody response than male participants at the follow-up visit. They also observed significant differences in vaccine antibody response across different age categories. The youngest participants had the highest antibody response at follow-up visit. Also, there is an insignificant general pattern of having a decreased antibody titer within older age groups.

Statistical Issues:

- The data set has several missing values. Especially, 90.1% of the participants completed a day 21 follow-up visit and 71.4% completed a month 6 follow-up visit. The missing values can appear due to other variables or unobserved data. For example, participants without antibody response or persistence may less likely complete the follow-up visit because they died of Ebola virus disease (EVD) prior to it. Therefore, the missing values can give a serious bias to the results in the paper.
- The baseline (day 0) issue should be scrutinized. They collected the data after the 30-min observation period postvaccination, and even some data were collected after 3 or 4 days postvaccination. Although some elevated baseline titers (above 607 EU/mL) were excluded from the analysis, this baseline can be an issue. If the baseline antibody titers were measured before vaccination, the values could be more reliable to compare with antibody titers at follow-up visit.
- As the authors said, the natural exposure can be the reason of the antibody response. Thus, their discussion of antibody titers cannot be used to draw conclusions regarding the protection conferred by vaccination.

2.

(a)

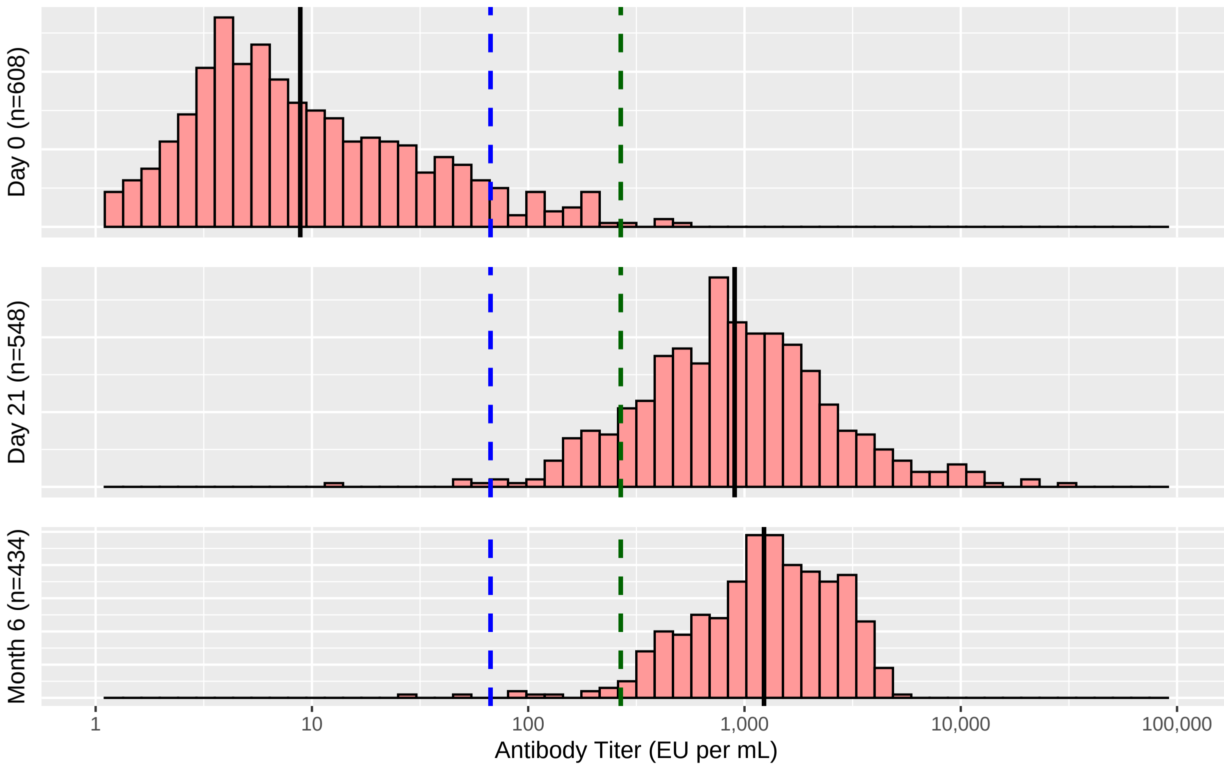

Figure 1: Histograms of antibody titers following rVSV∆G-ZEBOV-GP vaccination among 609 participants. The blue dashed line indicates 66.96 EU/mL, the LLOQ for FANG ELISA. The green dashed line indicates 267.84 EU/mL, four times the LLOQ for FANG ELISA. The black line indicates the geometric mean of antibody titers at each time point.

(b)

As mentioned in the summary, the missing values can affect on the results that 87.2% of samples had an antibody response at visit 2 and 95.6% demonstrated an antibody persistence at visit 3. However, Figure 1 suggests that there is a more serious issue about the numbers 87.2% and 95.6%.

The authors used not only the proportion between the antibody titers of visit 1 and the follow-up visit but also 4LLOQ, as the criteria for the antibody response and persistence. As shown in Figure 1, 10.9% of samples at day 21 are less than 4LLOQ, whereas only 2.76% at month 6 are less than 4LLOQ. Thus, it is likely that the number 87.2% extremely depends on the criterion 4LLOQ. If we use only the proportion criterion, 97.4% of the samples at day 21 have antibody response. Moreover, 92.3% have antibody response even when we define it as an antibody titer that had increased tenfold from visit 1 to visit 2. This value can also indicate the dramatically increased antibody levels. Therefore, using only proportion criterion would be better to determine the antibody response.

Admittedly, the 4LLOQ may be a standard criterion for the antibody response in the immunology field. However, this fixed constant can be adjusted because the baseline varies across the individuals. If we use 2LLOQ criterion and the proportion criterion (sixfold for tradeoff), 93.8% of the samples at day 21 have antibody response and 96.1% at month 6 have antibody persistence. These numbers may be more reliable.

3.

Since we are only given the median value, we cannot use the random effect of individuals. Thus, we can just consider OLS to fit the linear regression for `day21_level = log10(day21_FANG6pt_median)`. We also use log transformation for antibody levels at the other visit times. After removing all data including missing values, we first fit the regression with all covariates (`sex`, `day0_level`, `M6_level`, `BL_ever_cc_ebola`, `hcw_curr`, `age`, `vax_days`, `educ`, and `civ_stat`).

Table 1: Fitted coefficients of the linear regression model

	Estimate	Std. Error	Pr(>|t|)
(Intercept)	1.105	0.2684	4.745e-05
sex Female	-0.07892	0.04317	0.06833
day0_level	0.1288	0.03609	0.0004041
M6_level	0.6144	0.05754	2.082e-23
BL_ever_cc_ebola No	0.01027	0.1294	0.9368
BL_ever_cc_ebola Yes	0.006077	0.1322	0.9634
hcw_curr Yes	-0.06084	0.04202	0.1485
age	-0.003282	0.001892	0.08356
vax_days 1	-0.09815	0.07863	0.2127
vax_days 3	0.274	0.1298	0.03539
vax_days 4	0.7248	0.3811	0.05794
educ College/University or Graduate school	-0.04069	0.05051	0.4211
educ Finished secondary school	0.02634	0.04914	0.5923
educ None	-0.09766	0.1272	0.443
civ_stat Married/Cohabitating	-0.0212	0.1343	0.8747
civ_stat Single	-0.02219	0.1404	0.8745

There are some significant coefficients, such as those for sex, age, and antibody levels at day 0 and month 6. Now we do diagnostics for the model.

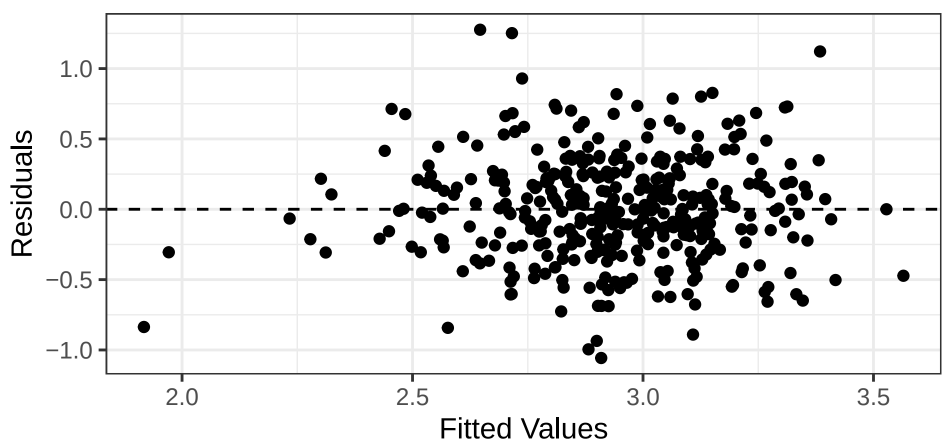

Figure 2: Fitted Values vs Residuals (the initial regression).

As shown in Figures 2 and 3, the residuals appear to have constant variance. Also, there is no independence or nonlinearity issue. Now we use backward elimination for model selection. At each step, it reduces a covariate which has the largest p-value of the F-test between the model at the step and the reduced model

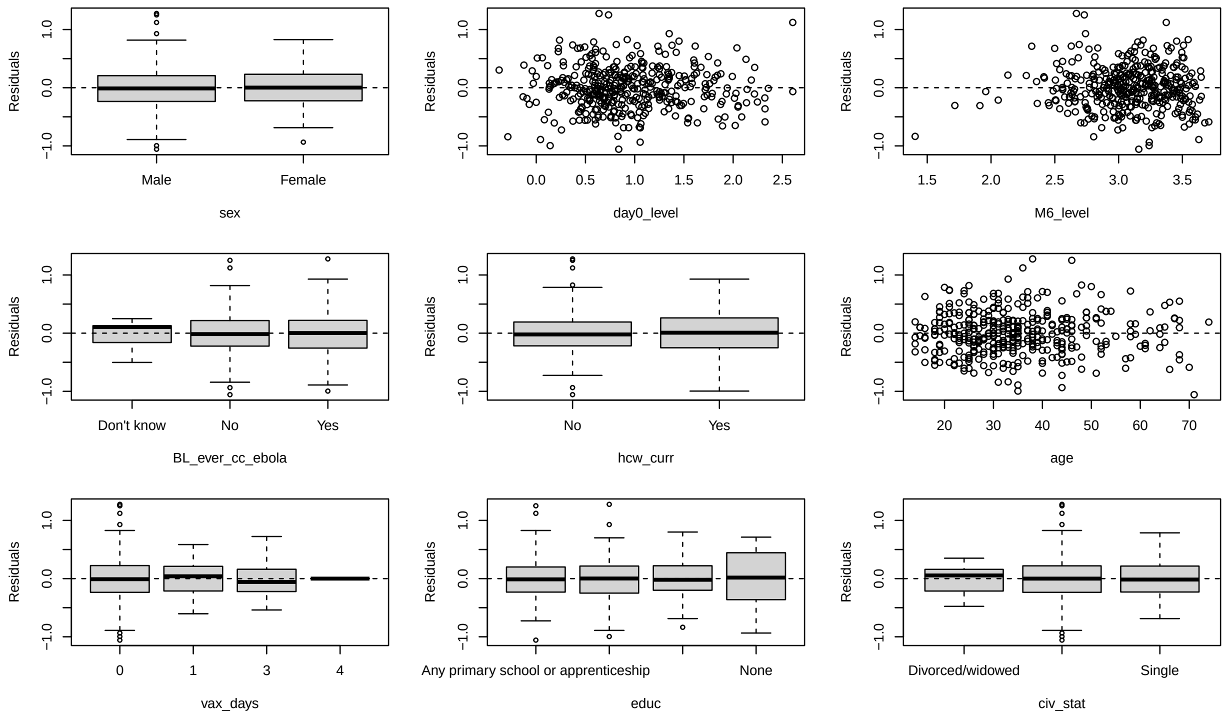

Figure 3: Each covariate vs Residuals (the initial regression).

without the covariate. When the largest p-value is less than 0.05, this process terminates. After the backward elimination, the covariates `day0_level`, `M6_level`, `age`, and `vax_days` are chosen. We fit the linear regression with these covariates.

Table 2: Fitted coefficients of the linear regression model with selected covariates.

	Estimate	Std. Error	Pr(>|t|)
(Intercept)	1.088	0.1874	1.35e-08
day0_level	0.1427	0.03515	5.967e-05
M6_level	0.5963	0.05643	4.582e-23
age	-0.003642	0.001505	0.016
vax_days 1	-0.09644	0.07789	0.2165
vax_days 3	0.2799	0.128	0.0294
vax_days 4	0.7317	0.3763	0.05255

Next, in order to consider interaction, we compare the fitted linear model to the augmented models with each interaction term. The p-values in Table 3 are larger than $0.00833 = 0.05/6$, so no interaction term is significant at the 5% level with Bonferroni correction.

Table 3: p-values of F-test between the fitted linear model and the augmented model with each interaction.

day0 : M6	day0 : age	day0: vax	M6 : age	M6 : vax	age : vax
0.0834	0.7146	0.4389	0.2308	0.478	0.2227

Therefore, our final linear model is

$$y_i = \beta_0 + \beta_{\text{day0\_level}} \text{ day0\_level}_i + \beta_{\text{M6\_level}} \text{ M6\_level}_i + \beta_{\text{age}} \text{ age}_i + \beta_{\text{vax\_days}} \text{ vax\_days}_i + \epsilon_i$$

and the fitted model has the coefficients in Table 2. As shown in Figure 4, the assumptions of constant variance, independence, and normality hold. Although there are very few potential outliers, we can conclude that this fitted model is appropriate.

Lastly, we interpret the fitted coefficients of the final model. The very significant positive coefficients for antibody titers at day 0 and month 6 indicate that there would be strong positive correlation between the antibody levels. It is natural because the antibody levels at day 21 would be affected by those at day 0, and the levels at month 6 would also be affected by those at day 21. The significant negative coefficient for age indicates that the antibody levels at day 21 would be a little higher for younger people. It corresponds to the result in the paper. The significant positive coefficients for `vax_days 3` and `vax_days 4` indicate that antibody levels would be higher for participants who got vaccination longer before.

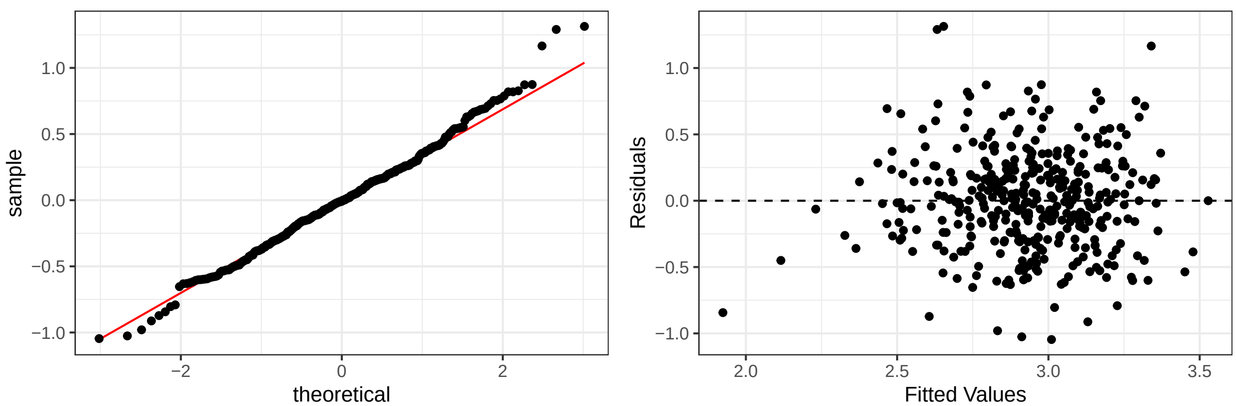

Figure 4: QQ plot of the residuals (left, the final model). Fitted values vs Residuals (right, the final model).

4.

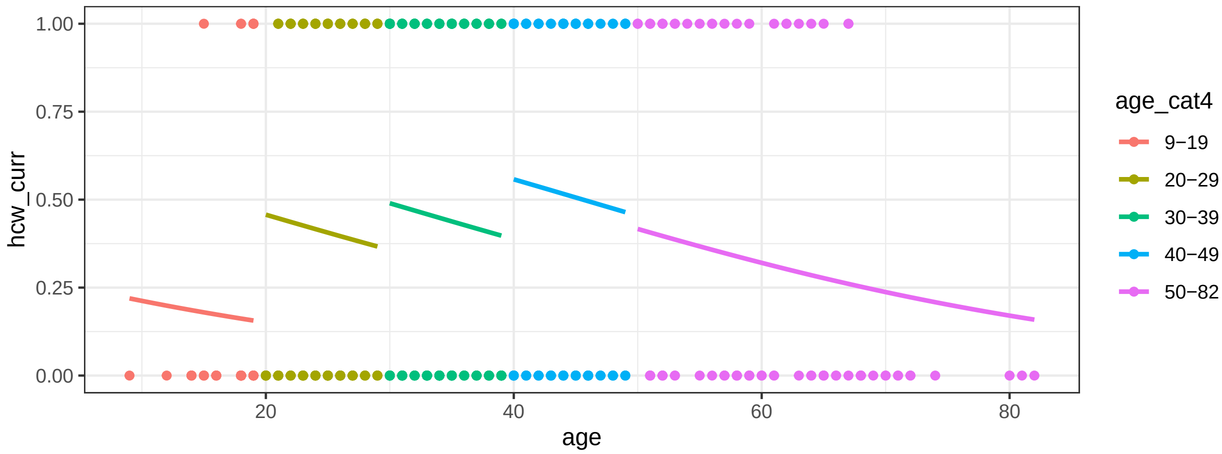

Figure 5: `hcw_curr` vs `age`. The curves are the fitted probabilities that the participant is a healthcare worker, by the model `glm(hcw_curr~age+as.factor(age_cat4), family=binomial(link = "logit"))`.

Using a continuous covariate and its categorical version is not natural. It is because there would be jumps at the border between the categories as we can see in Figure 5. It ignores the continuous properties. Moreover, the slopes are same in all categories. In this case, the fitted slope is negative since the decreasing trend after 40 is very strong. The increasing trend before 40 is only captured by increasing steps of intercepts for each category, while the each slope is negative. The both trends should be captured by the continuous covariate without any jump. One may consider adding the interaction term `age:age_cat4` for different slopes and intercepts. However, the categories may not be divided properly and the model with interaction term would generate jumps again. Therefore, we will use only the continuous covariate `age`.

We propose a generalized linear model with a quadratic term as an alternative model:

$$\text{logit}(p_t) = \beta_0 + \beta_1 t + \beta_2 t^2$$

where t is an age and p_t is the probability that the t-year-old participant is a healthcare worker.

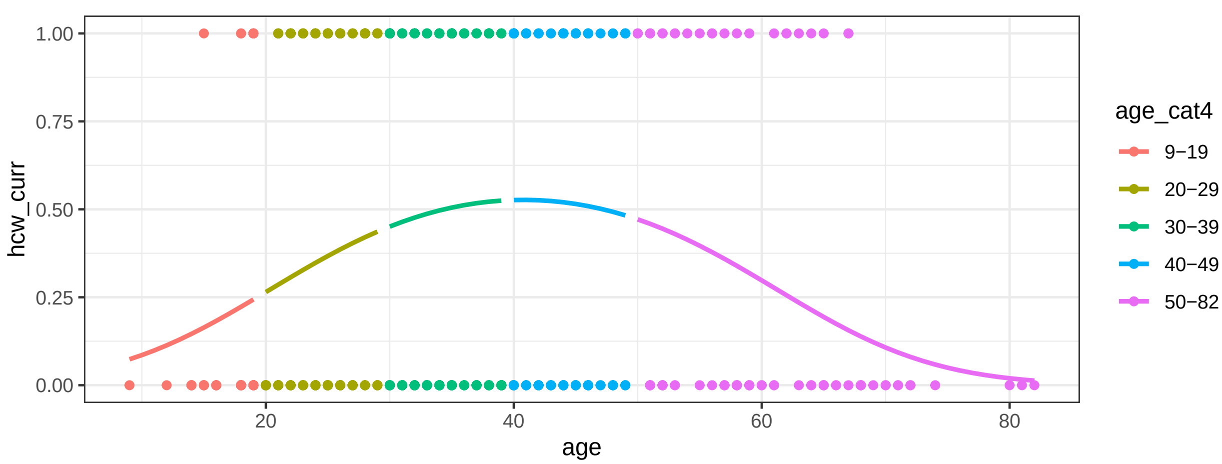

Figure 6: `hcw_curr` vs `age`. The curves are the fitted probabilities that the participant is a healthcare worker, by the alternative model.

Table 4: Fitted coefficients of the alternative model

	Estimate	Std. Error	Pr(>|z|)
β_0	-4.237	0.7106	2.49e-09
β_1	0.213	0.03802	2.123e-08
β_2	-0.002611	0.0004783	4.78e-08

The fitted coefficients of our alternative model are very significant. Also, the curve in Figure 6 seems very suitable without any jump. The increasing trend before 40 and the decreasing trend after 40 are captured properly. Hence, this model would be the most appropriate alternative model.

We can interpret the trends in practice. Too young or too old people may not tend to work and people between 30~50 ages would be most active to work in their life. Then, the relationship between `hcw_curr` and `age` would be like the curve in Figure 6. This type of relationship might lead to the fitted coefficients that we see in Figure 5 if we use the original model. On the contrary, our alternative model would capture this relationship correctly.

5.

(a)

We fit the linear mixed-effect model:

$$\log(y_{ij}) = X_i^\top \beta_j + \gamma_j + u_i + \epsilon_{ij}, \quad i = 1, \cdots, 609, \; j = 0, 1, 2$$

where γ_j is a fixed effect for visit (day 0, day 21, and month 6), $u_i \sim N(0, \sigma_u^2)$ is a random effect for individuals, $\epsilon_{ij} \sim N(0, \sigma_e^2)$ is an error term, and X_i is the vector of predictors including age, sex, marital status, health care workers status, vaccination days, education, and Ebola close contact or not. Note that we include the fixed effect γ_j for visit because it has only three levels and treating it as a random effect is therefore not appropriate. In addition, we set `NA` as a factor level for some covariates (`BL_ever_cc_ebola`, `hcw_curr`, `vax_days`, `educ`, and `civ_stat`) to perform imputation of the missing measurements. If fitting the model is not for imputation, we would just remove the data which include `NA` for some covariates.

Table 5: Fitted coefficients of the linear mixed-effect model (omitting the other coefficients of the interaction terms)

	Estimate	Std. Error	Pr(>|t|)
(Intercept)	2.704	0.4573	4.167e-09
age	-0.007154	0.004125	0.0831
sex Female	-0.3521	0.09279	0.000154
civ_stat Married/Cohabitating	-0.4162	0.2884	0.1491
civ_stat Single	-0.4791	0.3064	0.1181
civ_stat `NA`	-1.811	0.6989	0.009652
hcw_curr Yes	-0.03949	0.09483	0.6772
hcw_curr `NA`	2.686	0.8704	0.00207
vax_days 1	0.2971	0.2017	0.141
vax_days 3	0.003723	0.2584	0.9885
vax_days 4	-0.4957	1.061	0.6404
vax_days `NA`	-1.297	0.6163	0.03549
educ College/University or Graduate school	-0.3898	0.1113	0.0004744
educ Finished secondary school	-0.01526	0.111	0.8906
educ None	0.5095	0.2409	0.03462
educ `NA`	-1.047	1.375	0.4465
BL_ever_cc_ebola No	0.4044	0.3107	0.1932
BL_ever_cc_ebola Yes	0.3756	0.3159	0.2346
BL_ever_cc_ebola `NA`	0.4761	0.3344	0.1547
visit day 21	4.564	0.5981	5.679e-14
visit month 6	4.292	0.6579	1.062e-10
sex Female : visit month 6	0.6499	0.136	2.027e-06
educ None : visit day 21	-0.8399	0.3373	0.01293
σ_u^2	0.1738		
σ_e^2	0.9322		

As show in Table 5, there are some significant fitted coefficients, such as those for `age`, `sex`, and `visit`. The negative coefficients for `age` and `sex` may reflect some decreasing patterns between antibody titer and age, and female. The intercepts 4.564 and 4.292 for day 21 and month 6 reflect the dramatic increase in the antibody titers.

Based on the fitted coefficients, we fill the missing values $y_{i'j'}$ with $\exp\left(\hat{\mathbb{E}}[\log(y_{ij})]\right) = \exp\left(X_{i'}^\top \hat{\beta}_{j'} + \hat{\gamma}_{j'}\right)$. The top plot in Figure 7 presents the inputted values. The values for each visit are in each cluster which is also shown in Figure 1. Therefore, the imputation of the missing measurements seems reasonable.

On the other hand, we can also use the conditional mean $\hat{\mathbb{E}}[u_i \mid y]$ of the random effects u_i since we know the ID for each missing value. Thus, we can fill the missing values $y_{i'j'}$ with $\exp\left(X_{i'}^\top \hat{\beta}_{j'} + \hat{\gamma}_{j'} + \hat{\mathbb{E}}[u_{i'} \mid y]\right)$. The bottom plot in Figure 7 presents the inputted values. The values for each visit are also in each cluster. Moreover, the wider dispersion of the values indicates that the imputation with adding the conditional means reflects the (random) effect of the individuals.

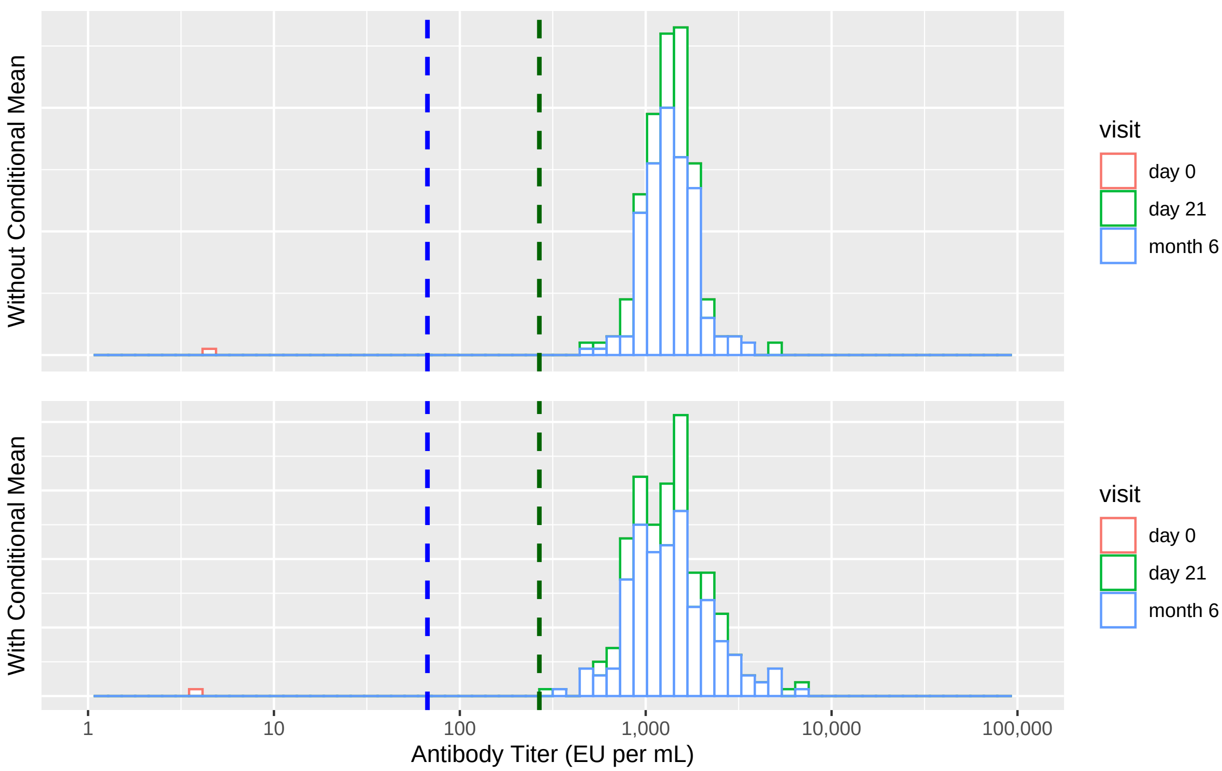

Figure 7: Histograms of inputted antibody titers. The blue and green dashed lines indicate LLOQ and 4LLOQ like in Figure 1.

(b)

The linear mixed-effect model $\log(y_{ij}) = X_i^\top \beta_j + \gamma_j + u_i + \epsilon_{ij}$ has several assumptions.

- The covariates are related linearly to the response, i.e., $\mathbb{E}[\log(y_{ij})] = X_i^\top \beta_j + \gamma_j$.
- The errors ϵ_{ij} are independent, and the random effects u_i are independent. Also, the errors ϵ_{ij} and u_i are independent.
- ϵ_{ij} has the constant variance σ_e^2. Also, u_i has the constant variance σ_u^2. Thus, we can expect the residual $u_i + \epsilon_{ij}$ would has the constant variance $\sigma_e^2 + \sigma_u^2$.
- ϵ_{ij} and u_i are normally distributed. Thus, we can expect the residual $u_i + \epsilon_{ij}$ would be normally distributed.

When we check several diagnostic plots, we find one serious plot. Figure 8 shows that there are some issues about constant variances and independence, while the left plot in Figure 9 shows that the normality assumption would hold. The variance is the largest at day 0 and it decreases at day 21 and more at month 6. We can also check this phenomenon in Figure 1. Thus, the model assumption about constant variance is violated. The most important effect of this heteroskedasticity is that the random effect would be underestimated. Since we added only intercept terms for the visit time, the random effects and the error terms should capture each variance. However, with the constant variance assumption, the fitted $\hat{\sigma}_u^2$ would be not sufficient for the data at day 0, even though it is sufficient for the data at month 6.

The correlation in each cluster in Figure 8 can also explained by this issue. The right plot in Figure 9 indicates that each u_i (conditional mean in the plot) should be correlated to the fitted values to capture the large variance. It may result in the correlation between the fitted values and the residuals.

If we want to handle the heteroskedasticity issue, we should add weights according to the variance of each clusters. The weighted linear mixed-effect model would be better to capture the random effect and the error term.

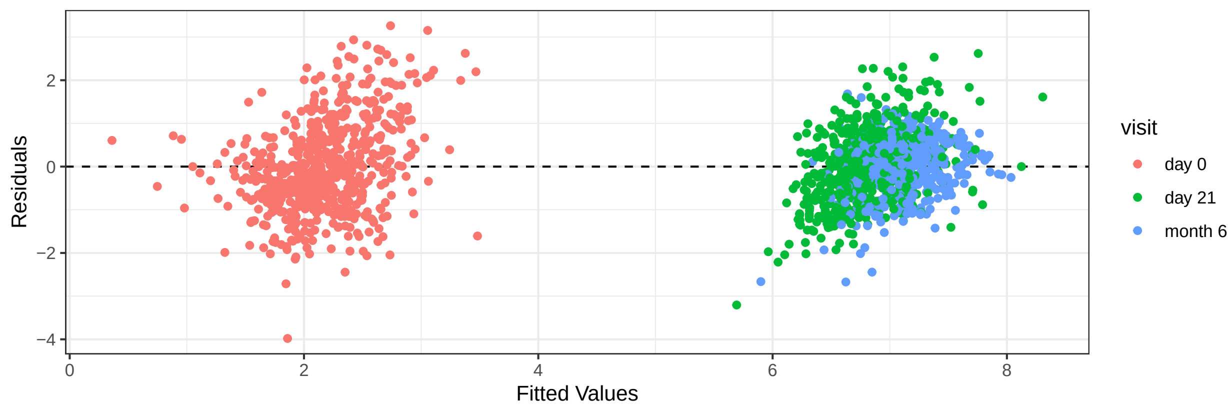

Figure 8: Fitted values vs Residuals of the linear mixed-effect model.

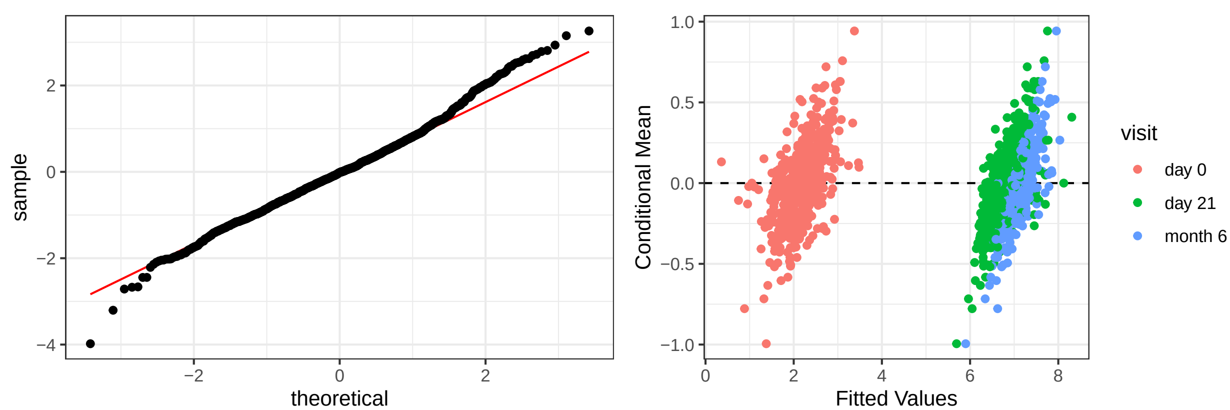

Figure 9: QQ plot of the residuals (left). Fitted values vs Conditional means of the random effects (right).

Lastly, the model only can capture the positive correlation between y_{ij} ($j = 0, 1, 2$) within an individual i. This is also a very serious issue. There can exist some negative correlation within each individual. Actually, the heteroskedasticity can be due to the negative correlation. Therefore, this model may be not reasonable to fit for this data. If we want to handle this issue, we should use the random effects for each visit time (i.e., using `(visit|ID)` instead of `(1|ID)` in the `lmer` code.) However, we cannot fit this larger model in our case because we have only one median measurement for each visit time and individual.

6.

We assume that the missing at random (MAR) assumption conditioning on covariates X_i holds: $y_i \perp\!\!\!\perp z_i \mid X_i$. Thus, $p(y_i \mid X_i)\, p(z_i \mid X_i) = p(y_i, z_i \mid X_i)$. We want to diagnose whether the missing completely at random (MCAR) assumption also holds or not. The MCAR assumption is $(X_i, y_i) \perp\!\!\!\perp z_i$, and it is equivalent to $p(X_i, y_i)\, p(z_i) = p(X_i, y_i, z_i)$. Since

$$p(y_i \mid X_i)\, p(X_i)\, p(z_i) = p(X_i, y_i)\, p(z_i) = p(X_i, y_i, z_i) = p(X_i)\, p(y_i, z_i \mid X_i) = p(X_i)\, p(y_i \mid X_i)\, p(z_i \mid X_i),$$

the MCAR assumption is equivalent to $p(z_i) = p(z_i|X_i)$, i.e., $z_i \perp\!\!\!\perp X_i$. Hence, we should check whether z_i and X_i are independent. Since we are only interested in X_i and z_i, we exclude y_i (`day21 FANG6pt median`). Also, we exclude several variables like `M6_FANG6pt_median`, `BL_ever_cc_ebola` because the features are measured at day 21 or month 6.

We use a permutation test to evaluate the independence. Let H_0 denote the null hypothesis that X_i and z_i are independent. Under H_0, for given $(X_1, z_1), (X_2, z_2), \cdots, (X_n, z_n)$,

$$T((X_1, z_1), (X_2, z_2), \cdots, (X_n, z_n)) \overset{d}{\equiv} T\left(\left(X_1, z_{\sigma(1)}\right), \left(X_2, z_{\sigma(2)}\right), \cdots, \left(X_n, z_{\sigma(n)}\right)\right)$$

for any permutation σ and any proper independence test statistic T. Then, for any t,

$$\mathbb{P}(T > t) \approx \frac{1}{B} \sum_{b=1}^{B} I_{(T_b > t)}$$

where $T_b = T\left(\left(X_1, z_{\sigma_b(1)}\right), \left(X_2, z_{\sigma_b(2)}\right), \cdots, \left(X_n, z_{\sigma_b(n)}\right)\right)$ for $b = 1, \cdots, B$. Thus, we can obtain the p-value of the statistic $T = T((X_1, z_1), (X_2, z_2), \cdots, (X_n, z_n))$ as $\frac{1}{B} \sum_{b=1}^{B} I_{(T_b > T)}$ approximately. If this p-value is significantly low, we can reject the null hypothesis and conclude that X_i and z_i are not independent.

In our case, under H_0, we would expect that the fitted coefficients are not significant when we fit the generalized linear model with the response z_i and the some covariates in X_i. Therefore, we use the absolute values of fitted coefficients as the statistic T, i.e., p-values would be $\frac{1}{B} \sum_{b=1}^{B} I_{\left(|\widehat{\beta}_b| > |\widehat{\beta}|\right)}$.

First, we fit the GLM model with the response z_i and predictors X_i including all features.

Table 6: The coefficients of the fitted GLM model

	Estimate	Std. Error	Pr(>|z|)
(Intercept)	15.18	623.8	0.9806
sex Female	0.5726	0.3197	0.0733
hcw_curr Yes	0.07993	0.3079	0.7952
age	0.02269	0.01435	0.1137
vax_days 1	1.203	1.033	0.2444
vax_days 3	-0.1669	0.7941	0.8336
vax_days 4	14.14	2400	0.9953
educ College/University or Graduate school	0.5264	0.3776	0.1633
educ Finished secondary school	0.2536	0.3491	0.4675
educ None	-1.195	0.5879	0.04208
civ_stat Married/Cohabitating	-14.33	623.8	0.9817
civ_stat Single	-14.17	623.8	0.9819
day0	0.09023	0.2555	0.724

The coefficients of `sex`, `age`, and `educ` are significant. These significant coefficients may already suggest that X_i and z_i are not independent. However, we cannot evaluate a p-value for H_0 and know whether this model is true or not, so we use the nonparametric test. Now we implement the permutation test by fitting GLM model with response z_i and the covariates `sex`, `age`, and `educ` for 2000 times.

Table 7: p-values of each coefficient of the GLM model

	p-value
(Intercept)	0.9925
sex Female	0.0425
age	0.0445
educ College/University or Graduate school	0.127
educ Finished secondary school	0.397
educ None	0.138

As shown in Table 7, the p-values for `sex` and `age` are significant at the 5% level. Figure 10 illustrates the p-values. Therefore, we can reject the null hypothesis and conclude that X_i and z_i are not independent. It means that it is unlikely the missingness pattern of antibody titers at day 21 is missing completely at random. The pattern would be missing at random only after conditioning on other features of the individuals. Based on the result in Table 6, we can conclude that male, younger people, or people not educated may have higher probabilities that they have missing values for antibody titers at day 21.

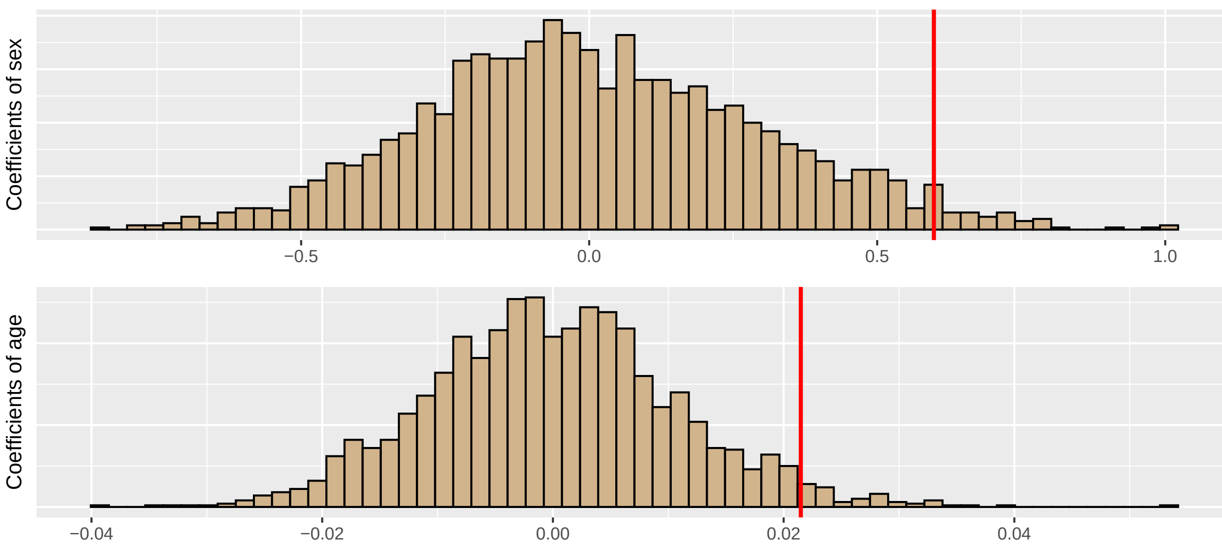

Figure 10: Coefficients of `age` and `sex` generated by permutation. The red lines indicate the fitted coefficients by the model `glm(z ~ sex + age + educ, family = binomial)` with the given data.

On the other hand, there is a limitation to our conclusion. The multiple testing issue can occur. If we already knew the best statistic T for the permutation test for H_0, we could just use it. However, we selected some significant covariates, implemented the permutation test, and compared each p-value to the 5% level. Thus, we need to use a global testing like Bonferroni correction taking into account the probabilities of all coefficients.

If we use Bonferroni test, we cannot reject the null hypothesis since the p-values in Table 7 are close to 5%. We can obtain the same conclusion when we just implement the permutation test using the fitted coefficients of the GLM model with the response z_i and predictors X_i including all features. Therefore, it is possible that the missingness pattern is missing completely at random.

Appendix

A. Question 2 Code

```
# library
library(readxl)
library(lmerTest)
library(dplyr)
library(tidyr)
library(ggplot2)
library(cowplot)
library(pander)

# load data
data = read_excel("pnas.2118895119.sd01.xlsx")

# treating as factor
data$sex = factor(data$sex, labels=c("Male", "Female"))
data$ID = as.factor(data$ID)
data$BL_ever_cc_ebola = factor(data$BL_ever_cc_ebola,
                               labels=c("Don't know","No", "Yes"))
data$day0_fang6pt = factor(data$day0_fang6pt, labels = c("No", "Yes"))
data$day21_fang6pt = factor(data$day21_fang6pt, labels = c("No", "Yes"))
data$M6_fang6pt = factor(data$M6_fang6pt, labels = c("No", "Yes"))
data$hcw_curr = factor(data$hcw_curr, labels = c("No", "Yes"))
data$age_cat4 = factor(data$age_cat4,
                       labels = c("9-19", "20-29", "30-39", "40-49", "50-82"))
data$vax_days = as.factor(data$vax_days)
data$educ = as.factor(data$educ)
data$civ_stat = as.factor(data$civ_stat)


# Figure 1
p1 = ggplot(data, aes(log10(day0_FANG6pt_median))) +
  geom_histogram(colour="black", fill="#FF9999", bins = 60) + xlim(0,5) +
  theme(axis.text.x=element_blank(),axis.ticks.x=element_blank(),
        axis.text.y=element_blank(),axis.ticks.y=element_blank(),
        axis.title.x = element_blank())+
  ylab("Day 0 (n=608)") +
  geom_vline(xintercept = mean(log10(data$day0_FANG6pt_median), na.rm=T), size = 1)+
  geom_vline(xintercept = log10(66.96), color="blue", size = 1, linetype = 2)+
  geom_vline(xintercept = log10(267.84), color="darkgreen", size = 1, linetype = 2)

p2 = ggplot(data, aes(log10(day21_FANG6pt_median))) +
  geom_histogram(colour="black", fill="#FF9999", bins = 60) + xlim(0,5) +
  theme(axis.text.x=element_blank(),axis.ticks.x=element_blank(),
        axis.text.y=element_blank(),axis.ticks.y=element_blank(),
        axis.title.x = element_blank())+
  ylab("Day 21 (n=548)") +
  geom_vline(xintercept = mean(log10(data$day21_FANG6pt_median), na.rm=T), size = 1)+
  geom_vline(xintercept = log10(66.96), color="blue", size = 1, linetype = 2)+
  geom_vline(xintercept = log10(267.84), color="darkgreen", size = 1, linetype = 2)
```

```r
p3 = ggplot(data, aes(log10(M6_FANG6pt_median))) +
  geom_histogram(colour="black", fill="#FF9999", bins = 60) +
  scale_x_continuous(breaks = 0:5, limits=c(0, 5),
                     labels=c("1", "10", "100", "1,000", "10,000", "100,000"))+
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank())+
  ylab("Month 6 (n=434)") + xlab("Antibody Titer (EU per mL)")+
  geom_vline(xintercept = mean(log10(data$M6_FANG6pt_median), na.rm=T), size = 1)+
  geom_vline(xintercept = log10(66.96), color="blue", size = 1, linetype = 2)+
  geom_vline(xintercept = log10(267.84), color="darkgreen", size = 1, linetype = 2)

plot_grid(p1,p2,p3, ncol = 1)


# check antibody titers less than 4LLOQ
mean(data$day21_FANG6pt_median <= 267.84, na.rm = T)
mean(data$M6_FANG6pt_median <= 267.84, na.rm = T)

# check antibody titers with a proportion criterion
mean((data$day21_FANG6pt_median/data$day0_FANG6pt_median)>=4, na.rm = T)
mean((data$day21_FANG6pt_median/data$day0_FANG6pt_median)>=10, na.rm = T)

# check antibody titers with a proportion criterion and 2LLOQ
mean((data$day21_FANG6pt_median/data$day0_FANG6pt_median)>=6 &
      data$day21_FANG6pt_median > 133.92, na.rm = T)
mean((data$M6_FANG6pt_median/data$day0_FANG6pt_median)>=6 &
      data$M6_FANG6pt_median > 133.92, na.rm = T)
```

B. Question 3 Code

```r
# data processing
data3 = data[,-c(3,4,7,8,9,12)]
data3$day21_level = log10(data$day21_FANG6pt_median)
data3$day0_FANG6pt_median = log10(data3$day0_FANG6pt_median)
data3$M6_FANG6pt_median = log10(data3$M6_FANG6pt_median)
colnames(data3)[2:3] = c("day0_level", "M6_level")
variables = colnames(data3)
data3 = na.exclude(data3)


# fit initial regression
fit = lm(day21_level ~ ., data3)

# Table 1
pander(summary(fit)$coefficients[,-3], align = "c")

# Figure 2
p = data.frame(fitted = fit$fit, resid = fit$res)
ggplot(p) + geom_hline(yintercept = 0, linetype = "dashed")+
  geom_point(aes(fitted, resid))+theme_bw()+
  xlab("Fitted Values") + ylab("Residuals")
```

```r
# Figure 3
par(mfrow = c(3,3),mar=c(4,4,2,2))
for(i in 1:9){
  p = data.frame(data3[,i], fit$res)
  plot(p, xlab = variables[i], ylab = "Residuals")
  abline(h = 0, lty = 2)
}


# backward_elimination
backward_elimination <- function(X){
  redX = X
  for(i in 1:10){
    n = length(colnames(redX))
    pval = rep(0,n-1)
    mod = lm(day21_level ~.,redX)

    #save p-value
    for(j in 1:(n-1)){
      Xtemp = data.frame(redX[,-j])
      pval[j] = anova(mod, lm(day21_level ~., Xtemp))$Pr[2]
    }

    #delete covariate
    if(sum(is.na(pval))==1){
      k = which(is.na(pval))
      redX = redX[,-k]
    }else if(sum(is.na(pval))>1){
      k= which(is.na(pval))[1]
      redX = redX[,-k]
    }else{
      k = which.max(pval)
      if(pval[k]>=0.05){
        redX = redX[,-k]
      }else{
        break
      }
    }
  }
  return(redX)
}
newdata = backward_elimination(data3)

# fit the final model
fit = lm(day21_level ~., newdata)

# Table 2
pander(summary(fit)$coefficients[,-3], align = "c")

# check interaction term
prob = c(anova(fit, lm(day21_level ~ . + day0_level:M6_level, newdata))$Pr[2],
         anova(fit, lm(day21_level ~ . + day0_level:age, newdata))$Pr[2],
         anova(fit, lm(day21_level ~ . + day0_level:vax_days, newdata))$Pr[2],
```

```
        anova(fit, lm(day21_level ~ . + M6_level:age, newdata))$Pr[2],
        anova(fit, lm(day21_level ~ . + M6_level:vax_days, newdata))$Pr[2],
        anova(fit, lm(day21_level ~ . + age:vax_days, newdata))$Pr[2]
)

# Figure 4
p = data.frame(fitted = fit$fit, resid = fit$res)
p1 = ggplot(p) + geom_qq_line(aes(sample = resid), col = "red")+
  geom_qq(aes(sample = resid)) + theme_bw()
p2 = ggplot(p) + geom_hline(yintercept = 0, linetype = "dashed")+
  geom_point(aes(fitted, resid))+theme_bw()+
  xlab("Fitted Values") + ylab("Residuals")
plot_grid(p1, p2)
```

C. Question 4 Code

```
data4 = data[!is.na(data$hcw_curr),]
x = 9:82
y = c(-0.89557 - 0.04157 * (9:19),
      -0.89557 + 1.55540 - 0.04157 * (20:29),
      -0.89557 + 2.10187 - 0.04157 * (30:39),
      -0.89557 + 2.79035 - 0.04157 * (40:49),
      -0.89557 + 2.63768 - 0.04157 * (50:82))
y = exp(y)/(1+exp(y))
z = factor(c(rep(1, 11), rep(2:4, each=10), rep(5, 33)),
           labels = c("9-19", "20-29", "30-39", "40-49", "50-82"))
w = data.frame(age=x, hcw_curr=y, age_cat4 = z)

# Figure 5
ggplot(data4, aes(age, as.numeric(hcw_curr)-1, col = age_cat4)) +
  geom_point() + theme_bw()+ ylab("hcw_curr") +
  geom_line(aes(age, hcw_curr,col = age_cat4),w, size=1)

# alternative model
data4$age2 = data4$age^2
fit = glm(hcw_curr~age+age2,data4,family=binomial(link = "logit"))

y = -4.2365561 + 0.2130010 * (9:82) -0.0026111 *(9:82)^2
w$hcw_curr = exp(y)/(1+exp(y))

# Figure 6
ggplot(data4, aes(age, as.numeric(hcw_curr)-1, col = age_cat4)) +
  geom_point() + theme_bw()+ ylab("hcw_curr") +
  geom_line(aes(age, hcw_curr),w, size=1)

# Table 4
pander(summary(fit)$coefficients[,-3], align='c')
```

D. Question 5 Code

```
# data processing
data5 = pivot_longer(data, cols = c(2,4,5), names_to = "visit", values_to = "titer")
data5$visit = factor(data5$visit, labels=c("day 0", "day 21", "month 6"))
data5$BL_ever_cc_ebola = factor(data5$BL_ever_cc_ebola, exclude = NULL)
data5$hcw_curr = factor(data5$hcw_curr, exclude = NULL)
data5$vax_days = factor(data5$vax_days, exclude = NULL)
data5$educ = factor(data5$educ, exclude = NULL)
data5$civ_stat = factor(data5$civ_stat, exclude = NULL)

# fitting the linear mixed-effect model
fit = lmer(log(titer) ~ (age + sex + civ_stat + hcw_curr +
             vax_days + educ + BL_ever_cc_ebola)*visit + (1|ID), data5)

# Table 5
pander(summary(fit)$coefficients[,-c(3,4)], align = "c")


# imputation
ind = is.na(data5$titer)
data5_1 = data5
data5_1$titer[ind] = exp(predict(fit, data5_1[ind,]))
data5_2 = data5
data5_2$titer[ind] = exp(predict(fit, data5_2[ind,]) +
                           unlist(ranef(fit))[data5_2$ID[ind]])

# Figure 7
p1 = ggplot(data5_1[ind,], aes(log10(titer), col = visit)) +
  geom_histogram(fill = "white", bins = 70) + xlim(0,5) +
  scale_x_continuous(breaks = 0:5, limits=c(0, 5),
                     labels=c("1", "10", "100", "1,000", "10,000", "100,000"))+
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank(),
        axis.text.x=element_blank(),axis.ticks.x=element_blank(),
        axis.title.x = element_blank())+
  ylab("Without Conditional Mean") +
  geom_vline(xintercept = log10(66.96), color="blue", size = 1, linetype = 2)+
  geom_vline(xintercept = log10(267.84), color="darkgreen", size = 1, linetype = 2)

p2 = ggplot(data5_2[ind,], aes(log10(titer), col = visit)) +
  geom_histogram(fill = "white", bins = 70) + xlim(0,5) +
  scale_x_continuous(breaks = 0:5, limits=c(0, 5),
                     labels=c("1", "10", "100", "1,000", "10,000", "100,000"))+
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank())+
  ylab("With Conditional Mean") + xlab("Antibody Titer (EU per mL)")+
  geom_vline(xintercept = log10(66.96), color="blue", size = 1, linetype = 2)+
  geom_vline(xintercept = log10(267.84), color="darkgreen", size = 1, linetype = 2)

plot_grid(p1,p2, ncol = 1)


# diagnostic
fit_df = fit@frame
```

```
fit_df$resid = residuals(fit)
fit_df$fitted = predict(fit, fit_df)
fit_df$cond = unlist(ranef(fit))[fit_df$ID]

# Figure 8
ggplot(fit_df) + geom_hline(yintercept = 0, linetype = "dashed")+
  geom_point(aes(fitted, resid, col = visit))+theme_bw()+
  xlab("Fitted Values") + ylab("Residuals")

# Figure 9
p1 = ggplot(fit_df) + geom_qq_line(aes(sample = resid), col = "red")+
  geom_qq(aes(sample = resid)) + theme_bw()
p2 = ggplot(fit_df) + geom_hline(yintercept = 0, linetype = "dashed")+
  geom_point(aes(fitted, cond, col = visit))+theme_bw()+
  xlab("Fitted Values") + ylab("Conditional Mean")
plot_grid(p1, p2)
```

E. Question 6 Code

```
# data processing
data6 = data[,c(1,10,11,13,14,15)]
data6$day0 = log10(data$day0_FANG6pt_median)
data6$z = factor(data$day21_fang6pt, labels = c(0,1))
data6 = drop_na(data6, vax_days, civ_stat, day0)

# Table 6
pander(summary(glm(z ~ ., data6, family = binomial))$coefficients[,-3])


# permutation test
a = summary(glm(z ~ sex + age + educ, data6, family = binomial))$coefficients
b = matrix(0, 2000, 6)
for(i in 1:2000){
  set.seed(i)
  newdata = data6
  newdata$z = sample(data6$z)
  b[i,] = summary(glm(z ~ sex + age + educ,
                      newdata, family = binomial))$coefficients[,1]
}

p = c()
for(j in 1:6){
  p[j]= mean(abs(b[,j])>abs(a[j,1]))
}
p = data.frame(p)
rownames(p) = rownames(a)
colnames(p) = "p-value"

# Table 7
pander(p)
```

```r
# Figure 10
g = data.frame(sex = b[,2], age = b[,3])

p1 = ggplot(g, aes(sex)) +
  geom_histogram(colour="black", fill="tan", bins = 60) +
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank(),
        axis.title.x = element_blank())+
  ylab("Coefficients of sex") +
  geom_vline(xintercept = a[2,1], col = "red", size = 1)

p2 = ggplot(g, aes(age)) +
  geom_histogram(colour="black", fill="tan", bins = 60)  +
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank(),
        axis.title.x = element_blank())+
  ylab("Coefficients of age") +
  geom_vline(xintercept = a[3,1], col = "red", size = 1)

plot_grid(p1,p2, ncol = 1)
```